Data Management and Sharing Plan

Products and Data Types

The project will produce source code, schemas, tests, documentation, containers and Spack recipes, software bills of materials (SBOMs), benchmark results, tutorial notebooks, synthetic datasets, Frozen Legacies catalog and dataset manifests, observation/flight-line GeoJSON, configuration/provenance manifests, and de-identified service-usage aggregates. Science demonstrations will use model inputs/outputs and publicly available observational products. Individual experiment outputs may be large and remain in user-owned project or HPC storage; the project will preserve the smaller inputs, configurations, diagnostics, and metadata required to understand and reproduce published results. The project will not collect sensitive human-subject research data.

Formats, Metadata, and Quality Control

Source and documentation will use Python, shell, YAML, JSON/JSON Schema, Markdown/MyST, Jupyter notebooks, and LaTeX. Scientific arrays will use NetCDF/HDF5 or Zarr with CF-style variable names, dimensions, coordinates, units, fill values, and attributes where applicable. Tabular aggregates will use CSV with a data dictionary. Each release will record semantic version, source commit, dependency locks, image digest, SBOM, supported platform, license, test results, and known limitations. Each qualified historical-data collection will record source, steward, license, attribution, spatial/temporal coverage, coordinate reference system, file inventory and checksums, transformations, uncertainty, and expert-reviewed quality-control status. Each reproducible run bundle will record application, dataset-adapter/model version, observation source and checksum, configuration, random seeds, resource request, backend, timestamps, environment reference, output checksums, and diagnostic definitions.

Automated unit, schema, regression, numerical, restart, end-to-end, and portability tests will gate releases. Synthetic test data will be generated from documented scripts with fixed seeds. Scientific acceptance tolerances and reference outputs will be versioned and reviewed by the science lead.

Access, Licensing, and Reuse

Code, schemas, documentation, tests, and nonrestricted tutorial data will be released no later than the associated public release through the ICESEE-project GitHub organization. Tagged releases will be archived in Zenodo or an equivalent long-term repository with DOI and CITATION.cff metadata. The project will harmonize currently inconsistent BSD/MIT metadata in Year 1. The intended default for original CryoStack/ICESEE code is the permissive BSD 2-Clause License, subject to Georgia Tech approval, because it supports academic and commercial reuse while preserving attribution. Documentation and tutorial prose will use CC BY 4.0 where institutionally permitted. Third-party models, data, MATLAB components, and dependencies retain their own licenses; the distribution will include a machine-readable dependency/license manifest and will not redistribute restricted inputs.

Observation bundles and Frozen Legacies records will cite the original DOI/URL or archive identifier and preserve source license, required attribution, collection lineage, and documented transformations. Data too large or restricted for redistribution will be represented by a fetch script, checksum, metadata record, and access instructions. Publications will link to the exact release and archived artifacts needed to support their conclusions.

Usage, Privacy, and Security

Operational records will include a random experiment identifier, application, adapter/backend versions, coarse resource request, status, timestamps, and errors needed for reliability and metrics. Credentials, private keys, passwords, and access tokens will not be stored in the research dataset or public metrics. Account

email and optional affiliation will be separated from run telemetry with access limited to authorized project personnel. Public reports will use aggregate counts, suppress small demographic cells, exclude automated tests, and document definitions. Raw identifiable operational logs will be retained only as long as required for service security and annual report verification, then deleted according to a written retention schedule.

Preservation and Responsibilities

GitHub provides active version control, issue history, and review records; Zenodo provides archival releases and persistent identifiers. Annual metric extracts, analysis scripts, data dictionaries, benchmark reports, and final governance/operations documents will be deposited by the end of each reporting year. Georgia Tech systems will back up the production database and deployment configuration; restoration will be tested annually. The PI is responsible for software releases, provenance, security-sensitive access, and repository preservation. The Co-PI is responsible for scientific-data selection, scientific validation criteria, and citation completeness. Adapter maintainers are responsible for model-specific metadata and test fixtures. The team will review this plan annually and document material changes in project reports.